Pythagorean Theorem in Real-World Problems

Question Bank - Additional 6.04

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.

1. A window is 8 ft wide and 6 ft tall. How long is its diagonal?

2. An 11-ft ladder leans on a wall with its foot 4 ft from the wall. How high does it reach?

3. A ramp rises 3 ft over a run of 10 ft. How long is the ramp?

4. A field is 9 m by 12 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

5. Will a 15-in rod lie flat in a box 9 in by 11 in, corner to corner?

6. A kite's 17-m string is held at a spot 8 m from the point straight below the kite. How high is the kite?

7. A wire runs from the top of a 9-m pole to a stake in the ground 12 m from the pole's base. How long is the wire?

Choose one.

8. A 10-ft ladder reaches 8 ft up a wall. How far from the wall is its foot?

- ○ A. 6 ft
- ○ B. 2 ft
- ○ C. 36 ft
- ○ D. 12.8 ft

9. A wire runs from the top of a pole to a stake in the ground. Which length is the hypotenuse?

- ○ A. the pole
- ○ B. the wire
- ○ C. the ground from the pole to the stake
- ○ D. none of them